15E 1 SIMPLE CHROMATOGRAPHY

SPECIFICATION REFERENCE
15.21 15.22

LEARNING OBJECTIVES

- Know that chromatography separates components of a mixture using a mobile phase and a stationary phase.
- Be able to calculate R_f values from one-way chromatograms in paper and thin-layer chromatography (TLC) and understand the reasons for differences in R_f values.

PAPER CHROMATOGRAPHY

You may know about paper chromatography from your early experiences of chemistry. This is a suitable place to remind you of this technique and to explain some of the principles that relate to most types of chromatography. The word 'chromatography' comes from the Greek for colour (*chroma*) and writing (*graphein*) because mixtures of coloured substances were used in early experiments. However, the technique can also be used with colourless substances because these can be detected in other ways without considering their colours. Chromatography is used:

- to separate a mixture into its individual components
- to identify the components of a mixture by considering how far they have travelled up the paper.

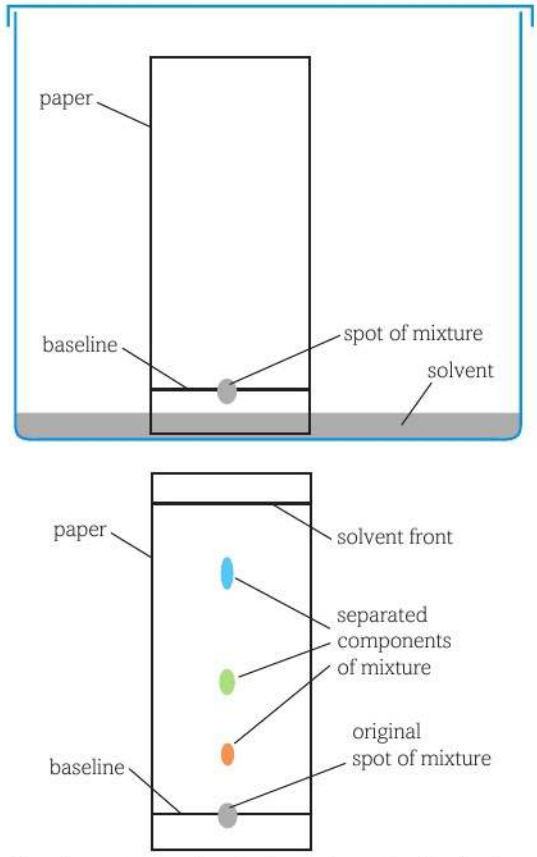
If you have synthesised an organic compound, chromatography is a good way to check whether it is pure and, if not, what impurities are present.

APPARATUS

All forms of paper chromatography need the following:

- A container (usually glass) with a lid. The lid is there to prevent evaporation of the solvent. The container could be a beaker or a rectangular tank.
- Paper. In introductory experiments you may have used filter paper or kitchen towel, but specially formulated chromatography paper gives better results. There also needs to be a method of supporting the paper in the container.
- A solvent. In some cases, water works well, but often the solvent is a mixture of organic compounds, and is chosen to fit the characteristics of the components of the mixture.

In paper chromatography, the mixture is spotted onto the paper a short distance from one edge. Often, spots of known substances are spotted separately at the same level as the mixture. A small volume of the chosen solvent is added to the container, and then the paper is inserted and suitably supported. The lid is replaced and the apparatus left until the solvent front has reached almost to the top of the paper. The paper is then removed, the position of the solvent front marked, and the paper left to dry. The resulting dried paper is known as a chromatogram. **Fig A** shows the typical apparatus used to set up a chromatography experiment and the resulting chromatogram.



▲ **fig A** The chromatogram here is shown in a rather idealised way – in reality the spots are often elongated and their centres are difficult to identify.

HOW DOES CHROMATOGRAPHY WORK?

All types of chromatography depend on the use of a **stationary phase** and a **mobile phase**. Each component in the mixture is attracted to both phases, but more strongly to one than the other.

- A component that is strongly attracted to the stationary phase but weakly attracted to the mobile phase will not travel very far up the paper.
- A component that is weakly attracted to the stationary phase but strongly attracted to the mobile phase will travel a long way up the paper.

In paper chromatography, the stationary phase is the water trapped in the fibres of the chromatography paper, and the mobile phase is the solvent.

We have used colours to illustrate the different substances in the mixture, but most organic compounds are colourless. Chromatography can still be used, but the components on the chromatogram must be visualised using ultraviolet radiation or by spraying with a chemical reagent that will react with the component to form a coloured product.

THIN-LAYER CHROMATOGRAPHY

The apparatus and method used in thin-layer chromatography (TLC) are very similar to those of paper chromatography. The only difference is that instead of paper, a sheet of glass or plastic coated in a thin layer of a solid such as silica or alumina is used (**fig B**).



▲ **fig B** This is a TLC plate after a chromatography experiment.

CALCULATING R_f VALUES

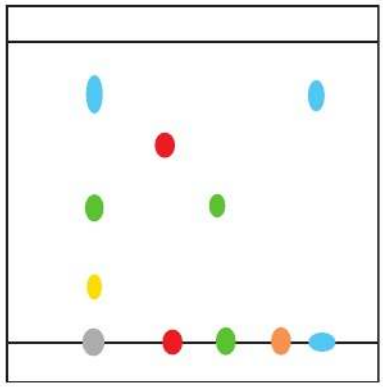
The chromatogram is analysed and the distance travelled by the solvent (from the baseline to the solvent front) is measured. For each component in the mixture, the distance it has travelled (starting from the baseline) is measured.

The R_f value is calculated using the expression:

$$R_f = \frac{\text{distance travelled by component}}{\text{distance travelled by solvent}}$$

R_f values have no units because they are a ratio of two distances. The letters in the term refer to 'retardation' (sometimes 'retention') and 'factor'. Each component has a characteristic R_f value. This means that, in theory, different components can be identified by reference to a table of known R_f values. Unfortunately, R_f values depend on the solvent used and other factors.

In practice, paper and thin-layer chromatography are usually carried out with known substances included in the same experiment so that a comparison can be made.



▲ **fig C** Chromatogram.

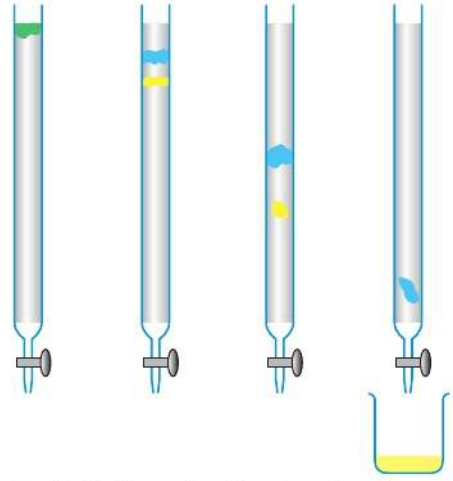
The chromatogram in **fig C** suggests that the mixture on the left contains three different substances. Two of these three are probably the green and blue substances on the right. The third substance (yellow) cannot be identified by this experiment.

LEARNING TIP

Remember, when calculating R_f values, the distance moved by the solvent is measured from the baseline to the solvent front.

COLUMN CHROMATOGRAPHY

Column chromatography uses the same principles as thin-layer chromatography. The stationary phase is alumina or silica packed into a tube (a burette will be suitable) and soaked in a solvent. The mixture is placed on top of the stationary phase and more solvent (the mobile phase) added on top. When the tap is opened, the solvent drips through the tip and the components of the mixture begin to move down the tube and separate.



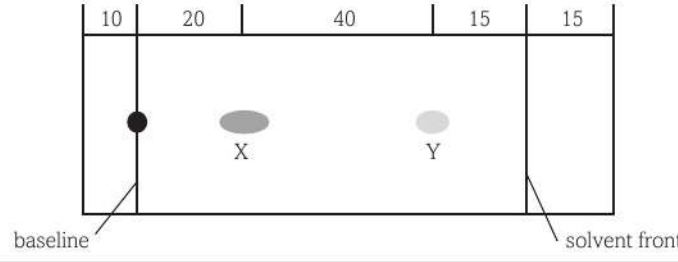
▲ **fig D** Separation of a mixture by column chromatography.

More solvent is added at the top, and eventually one component (yellow in this example) leaves the column and can be collected in a container (**fig D**). If the experiment is continued, the blue component can be collected in the same way.

The advantage of column chromatography is that much larger quantities of material can be separated than with paper chromatography.

CHECKPOINT

1. Suggest why, in paper chromatography, the non-polar substance hexane has a high R_f value.
2. Calculate the R_f value of component X in this diagram.



SUBJECT VOCABULARY

stationary phase in chromatography the liquid or solid that does not move

mobile phase in chromatography the liquid that moves through the stationary phase and transports the components

15E 2 DETERMINING STRUCTURES USING MASS SPECTRA

SPECIFICATION REFERENCE
15.17(i) 15.17(ii)

LEARNING OBJECTIVES

- Be able to use data from mass spectra to:
  - (i) suggest possible structures of a simple organic compound, given accurate relative molecular masses
  - (ii) calculate the accurate relative molecular mass of a compound, given accurate relative molecular masses to four decimal places.

MASS SPECTROMETRY SO FAR

In **Topic 10 (Book 1: IAS)** you learned about two aspects of mass spectrometry. They were:

- determining the relative atomic masses of elements from isotopic abundances
- using fragmentation patterns to determine the structures of organic compounds.

In this section we will learn about another use – determining the molecular formula (although not always the structure or identity) of an organic compound from a precise relative molecular mass obtained from **high resolution mass spectrometry** (HRMS).

HIGH RESOLUTION MASS SPECTROMETRY

Without going into technical detail, there are mass spectrometers that can provide a value for a relative molecular mass to four or more decimal places. Values with this degree of precision sometimes enable a compound to be positively identified from the relative molecular mass alone, without the need for any other information. Consider compounds with M_r = 58 (or 58.0). There are several possible structures, including those in **table A**:

FORMULA	STRUCTURE	ACCURATE M_r
C_4H_{10}	$CH_3CH_2CH_2CH_3$ or $(CH_3)_3CH$	58.0780
C_3H_6O	CH_3COCH_3 or CH_3CH_2CHO	58.0417
$C_2H_6N_2$	$H_2NCH{=}CHNH_2$	58.0530

table A

LEARNING TIP

HRMS gives information only about molecular formulae, not about structure. Structures can be deduced from the fragmentation patterns in mass spectrometry.

Once an accurate M_r has been obtained from HRMS, it is possible to decide whether the compound has the molecular formula C_4H_{10} or C_3H_6O or $C_2H_6N_2$. Note that if the accurate M_r is found to be 58.0417, it is still not possible to decide whether the compound is propanone or propanal because they both have the same accurate M_r value.

DID YOU KNOW?

Uses of HRMS include in forensics and environmental analysis.



▲ **fig A** HRMS equipment looks very similar to other mass spectrometry equipment.

CALCULATING THE ACCURATE RELATIVE MOLECULAR MASS

We can also obtain accurate M_r values by doing some simple calculations. You know that relative atomic masses are relative to the average mass of a carbon atom (^{12}C) taken as exactly 12, which for our purposes means 12.0000. **Table B** shows the accurate atomic masses of some common elements:

ELEMENT	SYMBOL	ACCURATE ATOMIC MASS
hydrogen	H	1.0078
carbon	C	12.0000
nitrogen	N	14.0031
oxygen	O	15.9949

table B

You will need to use information from **table B** in the following examples.

WORKED EXAMPLE 1

A compound is found to have an M_r value of 84.0573. Which of these structures does the compound have?

$CH_2{=}CHCH_2CH_2CH_2CH_3$ or $CH_3CH{=}CHCH_2CHO$

Answer

The molecular formula of $CH_2{=}CHCH_2CH_2CH_2CH_3$ is C_6H_{12} and its relative molecular mass is calculated as follows:

$(6 \times 12.0000) + (12 \times 1.0078) = 84.0936$

The molecular formula of $CH_3CH{=}CHCH_2CHO$ is C_5H_8O and its relative molecular mass is calculated as follows:

$(5 \times 12.0000) + (8 \times 1.0078) + 15.9949 = 84.0573$

This means that the compound is $CH_3CH{=}CHCH_2CHO$.

WORKED EXAMPLE 2

Three compounds have these structures:

A $CH_3CH_2CH_2CH{=}CH_2$
B $CH_3CH(NH_2)CN$
C $CH_2{=}CHCOCH_3$

In a high resolution mass spectrometer, compound Y is found to have an M_r value of 70.0423.

What is the identity of Y?

Answer

The molecular formula of A is C_5H_{10} and its M_r value is:

$(5 \times 12.0000) + (10 \times 1.0078) = 70.0780$

The molecular formula of B is $C_3H_6N_2$ and its M_r value is:

$(3 \times 12.0000) + (6 \times 1.0078) + (2 \times 14.0031) = 70.0530$

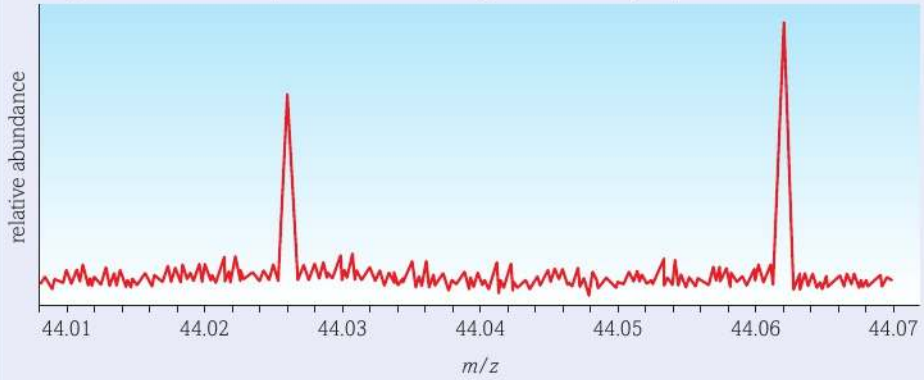
The molecular formula of C is C_4H_6O and its M_r value is:

$(4 \times 12.0000) + (6 \times 1.0078) + 15.9949 = 70.0417$

Although none of the calculated M_r values exactly matches the accurate M_r value obtained from the high resolution mass spectrometer, the value is much closer to C than to A or B, so compound Y is C.

WORKED EXAMPLE 3

The high resolution mass spectrum shown in **fig B** has two major peaks.



▲ **fig B** A high resolution mass spectrum.

Suggest which of these compounds are responsible for the mass spectrum:

P C_3H_8
Q CH_3CHO
R $O{=}C{=}O$

Answer

M_r of P = $(3 \times 12.0000) + (8 \times 1.0078) = 44.0624$

M_r of Q = $(2 \times 12.0000) + (4 \times 1.0078) + 15.9949 = 44.0261$

M_r of R = $12.0000 + (2 \times 15.9949) = 43.9898$

The two peaks correspond to P and Q, but not to R, so propane and ethanal are present, but not carbon dioxide.

SUBJECT VOCABULARY

high resolution mass spectrometry (HRMS) a type of mass spectrometry that can produce M_r values to several decimal places, usually four or more

CHECKPOINT

1. Two compounds have the structures $CH_3CH_2CH_2CH_2NH_2$ and HN=CHCOOH. In a high resolution mass spectrum, one of these compounds has M_r = 73.0812. Explain how you can decide which compound it is. You do not need to show any working.

2. A compound is thought to be either 1,2-diaminoethane or ethanoic acid. Its accurate M_r value is 60.0213. Which compound is it? Show your working.

SKILLS ANALYSIS

15E 3 CHROMATOGRAPHY: HPLC AND GC

SPECIFICATION REFERENCE 15.23 PART

LEARNING OBJECTIVES

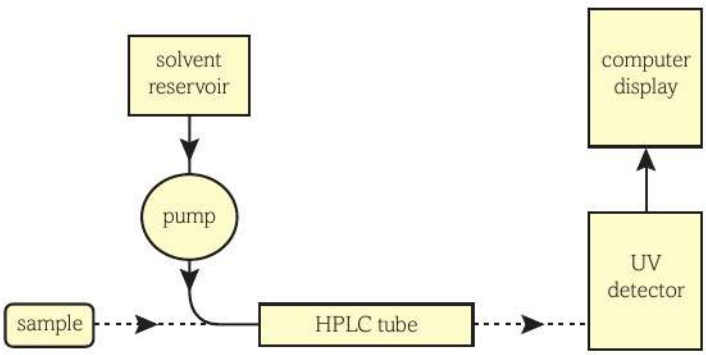
- Know that high performance liquid chromatography, HPLC, and gas chromatography, GC, are types of column chromatography that separate substances because of different retention times in the column.

HIGH PERFORMANCE LIQUID CHROMATOGRAPHY

High performance liquid chromatography (HPLC) is a refinement of column chromatography, described in **Section 15E.1**. The main differences are:

- The solvent is forced through a metal tube under high pressure, rather than being allowed to pass through by gravity.
- The particle size of the stationary phase is much smaller, which leads to better separation of the components.
- The sample is injected into the column.
- The components are detected after passing through the column, usually by their absorption of ultraviolet radiation.
- The whole process is automated, and the results are quickly available on a computer display.

Fig A shows a typical setup.



▲ **fig A** Apparatus for HPLC.

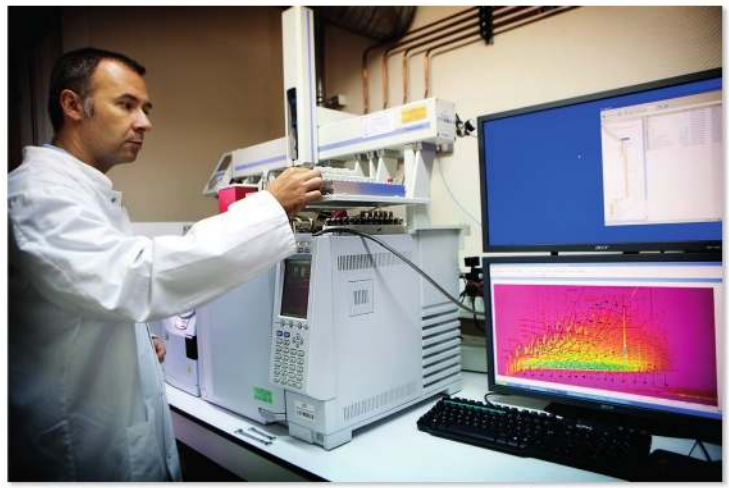
As each component reaches the detector, a signal is displayed on the computer screen. Just as retention factor (R_f) values can be obtained from paper and thin-layer chromatography, in HPLC values of **retention time** are obtained. The retention time of a component is the time taken from injection to detection, and these values can be important in identifying the components. However, retention times depend on several variables, including:

- the nature of the solvent
- the pressure used
- the temperature inside the column.

GAS CHROMATOGRAPHY

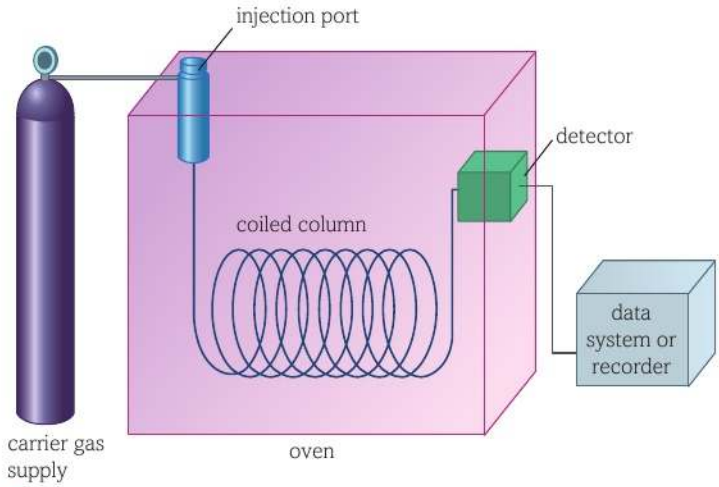
Gas chromatography (GC) is another refinement of column chromatography. The main differences are:

- The metal tube can be several metres long and is coiled to save space.
- The stationary phase is a solid or liquid coated on the inside of the tube.
- The mobile phase is an inert carrier gas (often nitrogen or helium).
- The sample is injected into the column, as in HPLC.
- The components passing through the column are detected.
- The whole process is automated, and the results are quickly available on a computer display (**fig B**).



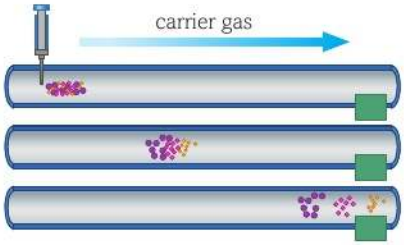
▲ **fig B** The complete GC equipment takes up very little room.

Fig C shows a typical setup.



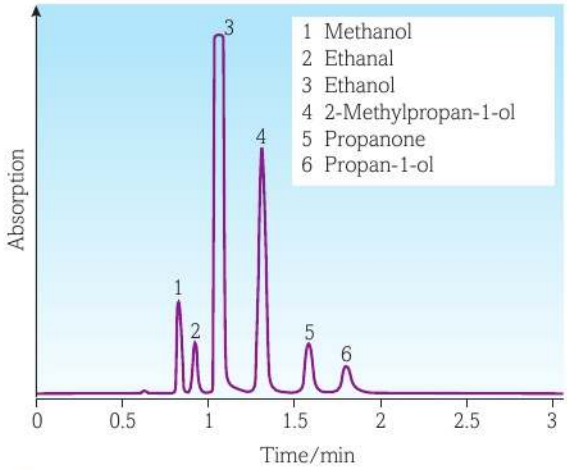
▲ **fig C** Apparatus for GC.

After the sample is injected, the components vaporise and move through the coiled tube with the carrier gas. They move at different speeds, depending on how strongly they are attracted to the stationary phase. Those with weaker attractions move more quickly and have shorter retention times. **Fig D** shows the separation occurring in the coiled tube.



▲ **fig D** A gas chromatograph separating three components.

The relative concentrations of the different components are often displayed on a graph like that in **fig E**.



▲ **fig E** The areas under the peaks represent the relative concentrations of the components.

LEARNING TIP

Traditional column chromatography (using a burette, for example) has one advantage over HPLC and GC – reasonable amounts of the components can be collected. In GC the components are often detected in a flame, so they are destroyed during the detection process at the end.

CHECKPOINT

1. How does the movement of the mobile phase in HPLC differ from that in traditional column chromatography?
2. Suggest reasons why retention times for a given substance may be different when obtained from different gas chromatograms.

SKILLS CRITICAL THINKING

SUBJECT VOCABULARY

retention time (of a component) in HPLC and GC, the time taken from injection to detection

15E 4 CHROMATOGRAPHY AND MASS SPECTROMETRY

SPECIFICATION REFERENCE 15.23 PART

LEARNING OBJECTIVES

- Know that high performance liquid chromatography, HPLC, and gas chromatography, GC, are types of column chromatography that separate substances because of different retention times in the column and may be used in conjunction with mass spectrometry, in applications such as forensics or drug testing in sport.

LIMITATIONS OF HPLC AND GC

The two types of chromatography described in **Section 15E.3** are very useful for separating small quantities of components in a mixture. However, they are not very good at positively identifying them. This is partly because it is difficult to control all of the variables (such as solvent, pressure and temperature) and partly because different substances may have the same retention times. Components are often identified by reference to a database of the retention times of known substances. However, if a component has a retention time for which there is no reference, then these techniques provide no useful information.

Here are two areas in which the results of these chromatography methods have to be exactly correct:

- in providing forensic evidence (for use in a court of law)
- in detecting banned drugs in sportsmen and sportswomen and racehorses.

They are used in other areas, including analysis of pollutants in the environment, detecting explosives in airport baggage and in space probes on other planets.

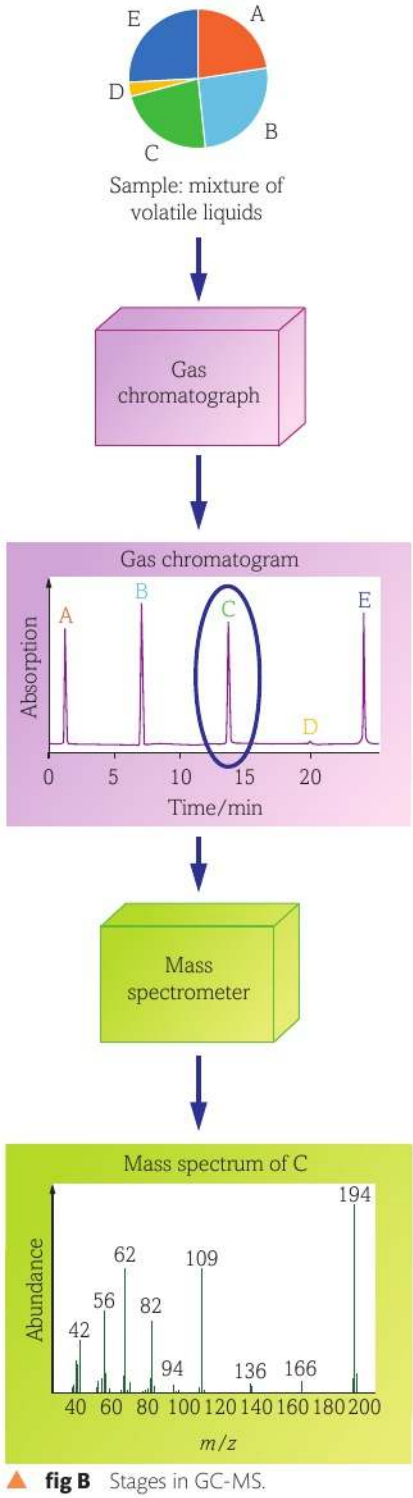
Because HPLC and GC do not give results that are beyond doubt, they are often combined with a different technique that you learned about in **Topics 2** and **10 (Book 1: IAS)**. This is mass spectrometry (MS), a technique that cannot separate a mixture but which can provide information about the structures of compounds. The combinations of these techniques are often abbreviated to HPLC-MS and GC-MS.



▲ **fig A** GC-MS is widely used for forensic purposes.

GC-MS

As the two combined techniques are very similar, we will focus only on GC-MS. **Fig B** shows the stages in a typical setup.



▲ **fig B** Stages in GC-MS.

Imagine a sample containing a mixture of five different substances, A–E.

Stage 1: The mixture is injected into a gas chromatograph.

Stage 2: Each component has a different retention time, so emerges at a different time.

Stage 3: One at a time, each component enters the mass spectrometer.

Stage 4: Component C has its mass spectrum displayed.

Stage 5: The m/z values and relative abundances of components are compared with a database of known substances.

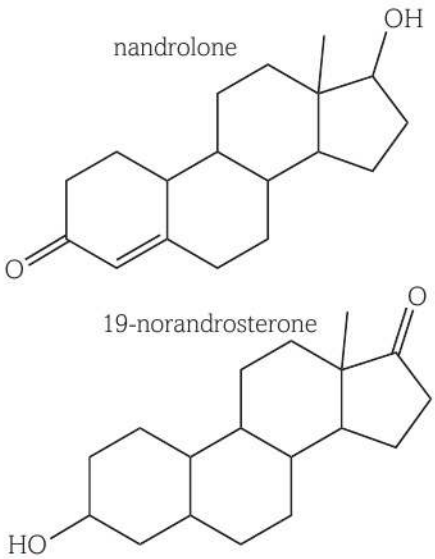
Stage 6: When a match is found, component C has been positively identified.

Very small traces of substances, for example in a human hair or a paint fleck, can be accurately identified by this method.

PROBLEMS WITH DRUG TESTING

Although the use of GC-MS for detecting small amounts of banned drugs is usually accepted as producing reliable results, sometimes a problem arises. One example involves the anabolic steroid called nandrolone. Anabolic steroids have legitimate uses in medicine, especially in the treatment of anaemia, osteoporosis and some forms of cancer. Its effects include muscle growth and increased red blood cell production and bone density.

Mainly because of their effects in promoting muscle growth, sporting authorities have prohibited the use of anabolic steroids in sportsmen and sportswomen, including athletes. When someone takes nandrolone, it is converted in the human body into a similar compound, 19-norandrosterone, which is then excreted in urine. Participants may be routinely tested for the presence of 19-norandrosterone in their urine, and the International Olympic Committee has set a limit of 2 nanograms (ng) per cm^3 of urine, which is only 0.000 000 002 g per cm^3. A participant with a concentration higher than this has 'tested positive'. The skeletal formulae of these two related compounds are:



▲ **fig C** Skeletal formulae of nandrolone and 19-norandrosterone.

In the 1998 Winter Olympic Games in Nagano, over 600 participants were tested for nandrolone, and none tested positive. In the years that followed, some very high-profile athletes in a range of sports have been banned from competition because of testing positive for nandrolone. However, many scientists believe that the number of positive tests is higher than expected. The suspicion is that nandrolone may be an ingredient in some nutritional supplements that participants are allowed to take. Another theory is that a combination of permitted dietary supplements and exercise may increase the concentration of nandrolone in the human body.



▲ **fig D** A urine sample from an athlete about to be tested for illegal drugs.

LEARNING TIP

In quoting very small concentrations of drugs, remember that a microgram (μg) is one millionth of a gram, 1×10^{-6} g.

CHECKPOINT

1. What is the main function of each technique in HPLC-MS?

SKILLS ETHICS

2. Which of the two compounds nandrolone and 19-norandrosterone would you expect to react with:

(a) bromine water

(b) 2,4-dinitrophenylhydrazine?

15E 5 PRINCIPLES OF NMR SPECTROSCOPY

SPECIFICATION REFERENCE
15.18 PART

LEARNING OBJECTIVES

- Understand the key aspects of NMR spectroscopy.
- Understand how ^{13}C and ^{1}H are detected by NMR.

EXAM HINT: EXTRA MATERIAL

You do not need to revise the principles of NMR spectroscopy, as you will not be asked questions about it in your exam.

WHAT IS NMR?

You learned about two major techniques used in analysing organic compounds in **Topic 10 (Book 1: IAS)**. These are mass spectrometry and infrared (IR) spectroscopy. In **Section 15E.2** you learned more about mass spectrometry (high resolution mass spectrometry).

This section and the following sections in **Topic 15** are about a third major technique, **nuclear magnetic resonance spectroscopy** (NMR). This is important for determining the structures of organic compounds. Every major hospital has a department with an NMR scanner, although they are normally described as MRI (magnetic resonance imaging) scanners.



▲ **fig A** A hospital patient about to have an MRI scan. MRI is a development of NMR spectroscopy.

A SIMPLIFIED EXPLANATION

NMR is much more difficult to fully understand than MS or IR spectroscopy. Fortunately, our emphasis will be on interpreting the spectra produced, and not on fully understanding how it works. Even so, it will help if you read a bit of the theory behind it. **Table A** shows some of the key points.

POINT	COMMENT
Nucleons (protons and neutrons) have spin.	You know that electrons in atoms that pair up in orbitals have opposite spins. Well, protons and neutrons in the nucleus also have opposite spins.
Nuclei have either an even or an odd number of nucleons.	In nuclei with an even number of nucleons, these spins cancel out. In nuclei with an odd number of nucleons, these spins cancel out, except for the odd one, which leaves the nucleus with residual spin.
Residual spin causes a tiny magnetic field.	Such nuclei can be imagined as tiny magnets.
These nuclei are affected by an external magnetic field.	Nuclei can be imagined as either lining up or opposing the external magnetic field.
There is a difference in energy between these two different states of a nucleus.	As the spins of nuclei 'flip' between lining up and opposing the external magnetic field, the nuclei can absorb electromagnetic radiation.
The nuclei of two adjacent (neighbouring) atoms in a molecule influence each other, and electrons also have an effect.	This means that electromagnetic radiation is absorbed differently by different atoms in a molecule, and these are detected as different 'signals' at different frequencies.

table A

KEY ASPECTS OF NMR

You need to know about several aspects of NMR.

NMR SPECTRUM

The output of a mass spectrometer or of an infrared spectrometer produces a spectrum (a kind of graph). Similarly, each compound tested using NMR produces a characteristic spectrum. The vertical axis is sometimes labelled 'absorption' (for 'absorption of radio frequency energy') and has no units, although the label is often omitted. The horizontal axis is labelled **chemical shift** (or sometimes with the δ symbol – the same symbol as used for partial charges) and has the units 'ppm' (parts per million). The scale usually starts with a zero value on the right, with the values increasing to the left.

WHICH ATOMS ARE DETECTED BY NMR?

For an atom to show up in an NMR spectrum, it must have an odd number of nucleons (protons and neutrons). That rules out carbon atoms, or at least ^{12}C atoms, as they each have three pairs

of protons and three pairs of neutrons. However, in naturally occurring carbon, about 1.1% of the carbon atoms are of the isotope ^{13}C, and they have an odd number of neutrons (seven). So, carbon atoms in molecules can be detected because some of them are ^{13}C atoms. This process is known as carbon-13 or **^{13}C NMR**. Hydrogen atoms are also detected – at least the ^{1}H isotopes that make up 99.985% of all hydrogen atoms. In the context of NMR, ^{1}H hydrogen atoms are often referred to as protons, so an alternative name for the process is **proton NMR**.

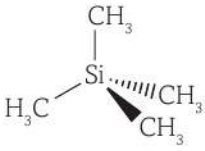
We often refer to these two atoms (^{13}C and ^{1}H) as 'producing a signal'. In this book we will consider only these two atoms, although NMR can be successfully used for other atoms, including ^{15}N and ^{31}P.

SUITABLE SOLVENTS

The organic compound being analysed has to be dissolved in a solvent, but most solvent molecules contain carbon and hydrogen atoms. This means that they will produce signals that interfere with those produced by the compound being analysed. The commonest solvent used has the formula $CDCl_3$. The symbol D stands for deuterium, which is the ^{2}H isotope of hydrogen. Deuterium atoms contain a proton and a neutron, and their spins cancel out. This solvent produces no signals that could interfere with the signals from hydrogen atoms. It produces only one signal that can interfere with the signals from carbon atoms, and this can easily be removed from the spectrum.

TMS

Carbon and hydrogen atoms in different chemical environments will produce different signals. Therefore, there needs to be a reference standard that will give a prominent signal. The compound used for this purpose is tetramethylsilane (usually abbreviated to TMS). Its structure is shown below:





▲ **fig B** The largest part of NMR equipment is often the large vacuum flask containing liquid helium, needed to cool the magnets to below 4 K.

You can see that it resembles a methane molecule, but with silicon instead of carbon and methyl groups instead of hydrogen atoms. SiH_4 is called silane, by comparison with methane, so perhaps you can understand the origin of the name tetramethylsilane. The molecule is suitable because it contains 12 hydrogen atoms, all joined in the same way in a symmetrical arrangement, so it produces a single strong signal that is easy to identify. It is also chemically unreactive, so will not react with most organic compounds.

CHEMICAL SHIFT

Each ^{13}C and ^{1}H atom in a compound produces a signal at a characteristic chemical shift (δ value), depending on the other atoms it is joined to. The reference standard, TMS, is given a chemical shift of zero (δ = 0.0 ppm), and all other atoms have values related to this. Chemical shift can be thought of as the resonant frequency of an atom. In **Sections 15E.6** and **15E.7** you will learn to use a data booklet containing values (or more often, ranges) of chemical shifts to identify specific atoms in molecules. Here are two examples:

carbon in a ketone group	—C—**C**O—C—	δ = 210 ppm
carbon in an alcohol	—**C**—OH	δ = 60 ppm

Note that these values refer only to the carbon atoms shown in bold, and not to any other carbon atoms or to any hydrogen atoms in the molecule.

LEARNING TIP

Do not worry if you think you do not understand the details of NMR theory – by far the most important thing for you to understand is its application.

CHECKPOINT

1. Some organic compounds contain these atoms: ^{16}O, ^{19}F, ^{32}S.
   Explain which of these are not suitable for use in NMR spectroscopy, and why.
2. Suggest why water (H_2O) is less suitable than TMS as a reference in NMR spectroscopy.

SUBJECT VOCABULARY

nuclear magnetic resonance spectroscopy (NMR) a technique used to find the structures of organic compounds. It depends on the ability of nuclei to resonate in a magnetic field

chemical shift (of a proton or group of protons) a number (in the units ppm) that indicates its behaviour in a magnetic field compared with tetramethylsilane. It can be used to identify the chemical environment of the carbon atoms or of the hydrogen atoms (protons) attached to it

^{13}C NMR the use of NMR spectroscopy to detect ^{13}C nuclei within the molecules of a substance, in order to determine the structure

proton NMR the use of NMR spectroscopy to detect ^{1}H nuclei within the molecules of a substance, in order to determine the structure

15E 6 ^{13}C NMR SPECTROSCOPY

SPECIFICATION REFERENCE
15.18 15.19(i) 15.19(ii)

LEARNING OBJECTIVES

- Understand that ^{13}C NMR spectroscopy provides information about the positions of ^{13}C atoms in a molecule.
- Be able to use data from ^{13}C NMR spectroscopy to:
  (i) predict the different environments for carbon atoms present in a molecule, given values of the chemical shift, δ
  (ii) justify the number of peaks present in a ^{13}C NMR spectrum in terms of the number of carbon atoms in different environments.

WHAT IS SHOWN BY A ^{13}C NMR SPECTRUM?

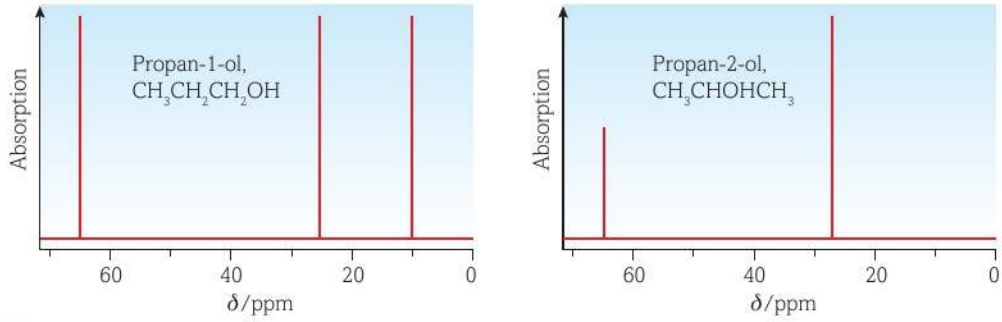
The actual spectrum of a compound is usually shown as a horizontal line just above the horizontal scale, but with vertical lines showing the signals produced by the ^{13}C carbon atoms in the molecule. These vertical lines are usually referred to as signals or peaks.

- The number of vertical lines tells you the number of different **chemical environments** of carbon atoms in the molecule, but not necessarily the total number of carbon atoms.
- The positions of the vertical lines on the horizontal scale tell you the chemical shifts of each carbon atom and, with reference to a list of chemical shifts, allow you to deduce the chemical environments.

The two isomers of propanol can be used to illustrate these points.

Propan-1-ol has the structure CH_3—CH_2—CH_2—OH. Each molecule contains three carbon atoms, all in different chemical environments. You might think that the two carbon atoms in the CH_2 groups are in the same chemical environment, but they are not. The second one is joined to CH_3 and CH_2, but the third one is joined to CH_2 and OH, which means they are in different chemical environments. There should therefore be three peaks in the spectrum.

Propan-2-ol has the structure CH_3—CH(OH)—CH_3. Each molecule contains three carbon atoms, but those in the two CH_3 groups are in the same chemical environment. This is because each CH_3 group is joined to CH(OH). This means that there should be two peaks in the spectrum. **Fig A** shows the ^{13}C spectra of these isomers.



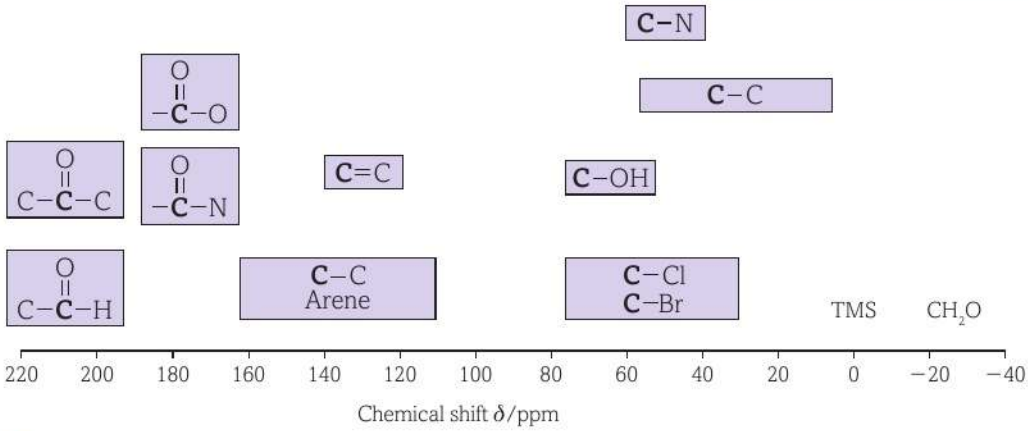
▲ **fig A** ^{13}C NMR spectra of propan-1-ol and propan-2-ol.

The spectra confirm the number of peaks expected, but show two other features.

- The less important feature is that in propan-1-ol the three peaks are of equal height, but in propan-2-ol the two peaks have different heights.
- The more important one is that the peaks have different chemical shifts.

INTERPRETING CHEMICAL SHIFTS

To interpret chemical shifts you need a table of values and corresponding carbon atoms or a chart. The chart in **fig B** will be provided in an examination, and you need to become familiar with using it.



▲ **fig B** ^{13}C NMR chemical shift ranges.

We will use this chart to analyse the two spectra for propanol.

Propan-1-ol:

In the spectrum of propan-1-ol the peaks appear at 65, 25 and 10 ppm. The only relevant ranges in the chart are C—C (the range is approximately 60–0 ppm) and C—OH (the range is approximately 75–55 ppm).

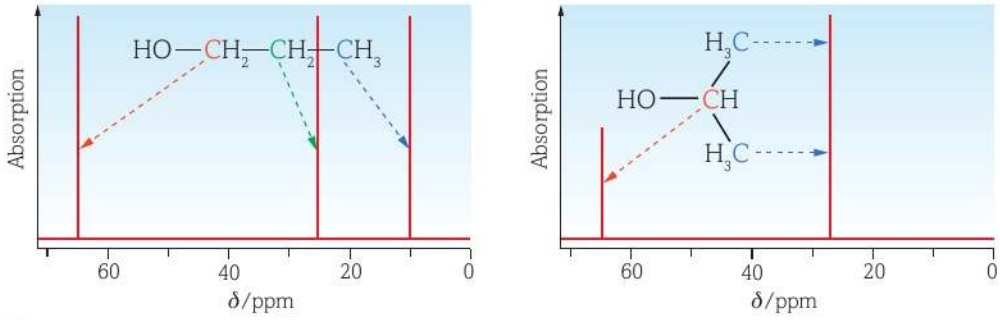
This means that the peak at 65 ppm corresponds to C—OH, so this is the carbon in the CH_2OH group. The peaks at 25 and 10 ppm correspond to C—C, so these are the carbons in the CH_3 and CH_2 groups. Notice that it is not possible to decide whether the peak at 25 ppm is due to the CH_3 or the CH_2 group.

Propan-2-ol:

In the spectrum of propan-2-ol the peaks appear at 65 and 27 ppm. The only relevant ranges in the chart are again C—C and C—OH.

This means that the peak at 65 ppm corresponds to C—OH, so this is the carbon in the CHOH group. The peak at 27 ppm corresponds to C—C, so this is due to the carbons in the two CH_3 groups. The peak at 27 ppm is bigger than the one at 65 ppm because it is caused by two carbon atoms, not one.

Fig C shows the interpretation of these two spectra.



▲ **fig C** Interpreting the ^{13}C NMR spectra of propan-1-ol and propan-2-ol.

PRACTICE MAKES PERFECT!

The way to check how well you understand how to use ^{13}C NMR spectra is to try lots of examples. In this section we will continue with some examples of varying types.

WORKED EXAMPLE 1

Predicting the number of peaks

You need to consider each carbon atom and whether it is uniquely positioned in the molecule or whether there is one or more carbon atoms identically positioned in the molecule.

For each of these compounds, predict the number of peaks due to the carbon atoms:

A $CH_3-CH_2-CH_2-CH_3$

B $(CH_3)_2-CH-CH_2-CH_3$

C $(CH_3)_4C$

The answers are in **table A**.

COMPOUND	NUMBER OF PEAKS	COMMENTS
A	2	One due to both CH_3 groups (these are both joined to $CH_2CH_2CH_3$, so are identical). One due to both CH_2 groups (these are both joined to CH_3 and CH_2CH_3, so are identical).
B	4	One due to the two left-hand CH_3 groups (these are identically joined to the rest of the molecule) – this should be bigger than the other two peaks because it is due to two carbon atoms. One due to the CH group. One due to the CH_2 group. One due to the right-hand CH_3 group (this is not identical to the other CH_3 groups because it is joined to CH_2 and they are joined to CH).
C	2	One due to the four CH_3 groups, which are all identically joined to the central C atom – this should be much bigger than the other peak as it is due to four carbon atoms. One due to the central C atom.

table A

WORKED EXAMPLE 2

Predicting the appearance of a spectrum

You need to consider all of the carbon atoms, as in Example 1, but this time also use the chart in **fig B** when considering the other atoms joined to them.

For each of these compounds, predict the number of peaks and chemical shift ranges due to the carbon atoms:

A $CH_3-CH_2-CH_2-NH_2$

B $(CH_3)_2CH-COOH$

C $(CH_3)_3N$

The answers are in **table B**.

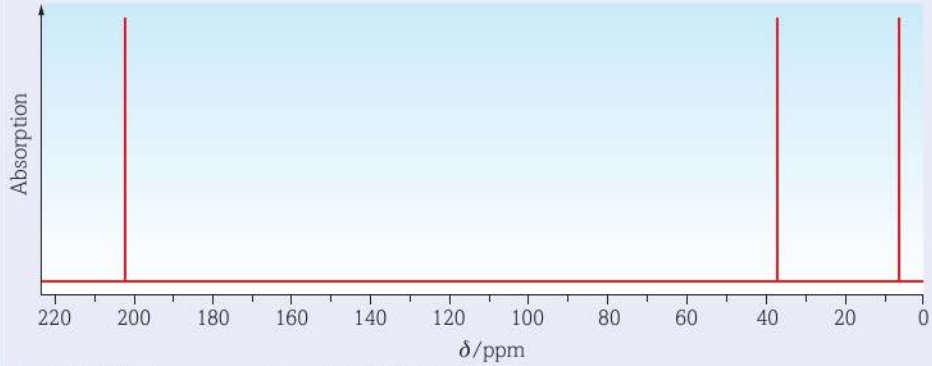
COMPOUND	NUMBER OF PEAKS	CHEMICAL SHIFT RANGES	COMMENTS
A	3	2 peaks in range 60–0 1 peak in range 63–35	Due to CH_3 and first CH_2 group. Due to CH_2 joined to NH_2.
B	3	1 peak in range 60–0 1 peak in range 60–0 1 peak in range 182–165	Due to the two CH_3 groups. Due to the CH group. Due to the COOH group.
C	1	1 peak in range 63–35	Due to the three CH_3 groups, which are all identical.

table B

WORKED EXAMPLE 3

Interpreting a spectrum

A compound has the molecular formula C_3H_6O and its ^{13}C NMR spectrum is shown in **fig D**.



▲ **fig D** Which compound has this ^{13}C NMR spectrum?

Use the spectrum and the chemical shift chart to explain whether the compound's structure is

$CH_2{=}CH{-}CH_2OH$ or $CH_3{-}CH_2{-}CHO$.

Answer

The peak at δ = 203 ppm is due to a carbon in an aldehyde or ketone.

The peaks at δ = 37 and 6 ppm are due to a carbon joined to another carbon by a single bond.

This interpretation fits $CH_3{-}CH_2{-}CHO$ but not $CH_2{=}CH{-}CH_2OH$.

If the structure were $CH_2{=}CH{-}CH_2OH$, there would have been a peak in the range δ = 140–115 ppm corresponding to C=C, and a peak in the range δ = 75–55 ppm corresponding to C–OH.

LEARNING TIP

When you need to decide how many different chemical environments there are for the carbon atoms in a molecule, it is a good idea to write the structure with the carbon atoms separated, e.g. $CH_3{-}CH(OH){-}CH_3$.

CHECKPOINT

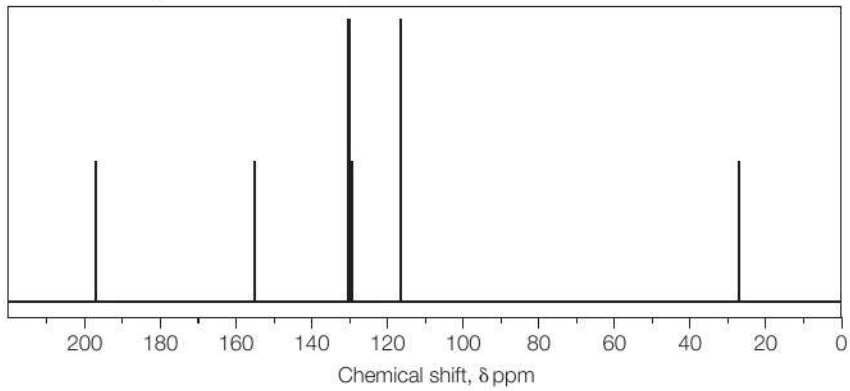
1. Work out the number of different chemical environments for the carbon atoms in the first six straight-chain alkanes.

SKILLS ANALYSIS

2. An aromatic compound has the molecular formula $C_8H_8O_2$.

   The compound contains both a phenol functional group and a ketone functional group.

   The ^{13}C NMR spectrum of the compound is shown below.



   Use the spectrum to deduce the structure of the compound. Explain your reasoning.

SUBJECT VOCABULARY

chemical environments the chemical environments of carbon atoms in a molecule are related to whether the carbon atoms are identically, or differently, positioned within a molecule

15E 7 ^{1}H NMR SPECTROSCOPY

SPECIFICATION REFERENCE
15.20(i) 15.20(ii)

LEARNING OBJECTIVES

- Be able to use both low and high resolution proton NMR spectroscopy to:
  (i) predict the different types of protons present in a molecule, given values of the chemical shift, δ
  (ii) relate relative peak areas, or ratio numbers of protons, to the relative numbers of ^{1}H atoms in different environments.

LOW RESOLUTION ^{1}H NMR SPECTROSCOPY

^{1}H NMR spectroscopy comes in two forms – low resolution and high resolution. The low resolution form is no longer used, but if we start with this, it will help you to understand the high resolution form more easily.

As with ^{13}C NMR spectroscopy you may see a peak at δ = 0.0 ppm. This is due to the reference standard tetramethylsilane (TMS) and should be ignored. The chemical shifts are measured relative to the peak for TMS.

WHAT DOES A LOW RESOLUTION ^{1}H NMR SPECTRUM SHOW?

The actual spectrum of a compound is usually shown as a horizontal line just above the horizontal scale, but with **peaks** produced by the ^{1}H hydrogen atoms in the molecule. The horizontal scale often starts from δ = 12.0 ppm on the left to 0.0 ppm on the right. Quite often, only the part of the scale where the peaks appear is shown.

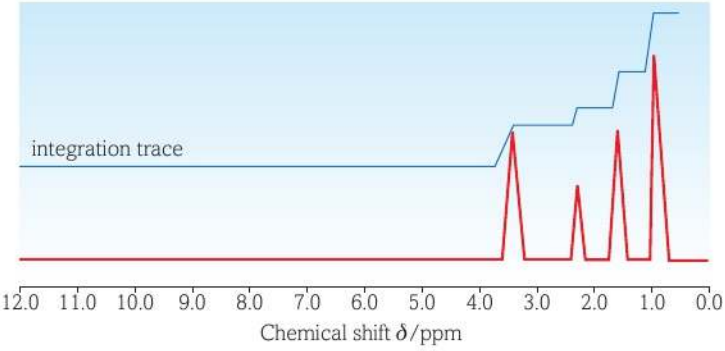
- The number of peaks tells you the number of different chemical environments of the hydrogen atoms in the molecule, but not the total number of hydrogen atoms.
- The positions of the peaks on the horizontal scale tell you the chemical shifts of each hydrogen atom, and, with reference to a list or chart of chemical shifts, these allow you to deduce the types of chemical environment.
- The areas under the peaks represent the relative numbers of hydrogen atoms in each environment.
- Sometimes a number is added at the side of each peak to show the relative area under the peak.

Another way to represent the relative areas is by an **integration trace**. This is a horizontal line that becomes higher as it passes each peak. The increases in height can be measured – they represent the relative areas under the peaks.

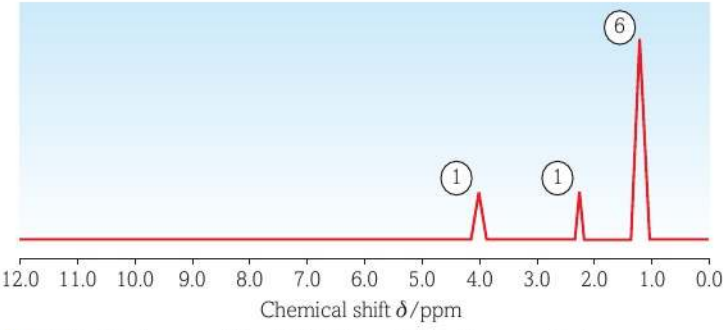
The two isomers of propanol can be used to illustrate these points.

Propan-1-ol has the structure CH_3–CH_2–CH_2–OH. There are eight hydrogen atoms, in four different chemical environments. You might think that the two pairs of hydrogen atoms in the CH_2 groups are in the same chemical environment, but they are not. The first CH_2 group is joined to CH_3 and CH_2, but the second one is joined to CH_2 and OH, and these are different chemical environments. This means that there should be four peaks in the spectrum. **Fig A** shows the low resolution ^{1}H NMR spectrum of this isomer with an integration trace.

Propan-2-ol has the structure CH_3–CH(OH)–CH_3. There are eight hydrogen atoms, but those in the two CH_3 groups are in the same chemical environments. This is because each CH_3 group is joined to $CH(OH)CH_3$. The hydrogen atoms in the CH group and in the OH group are in unique chemical environments. This means that there should be three peaks in the spectrum. **Fig B** shows the low resolution ^{1}H NMR spectrum of this isomer with the ratio of peak areas indicated by circled numbers.



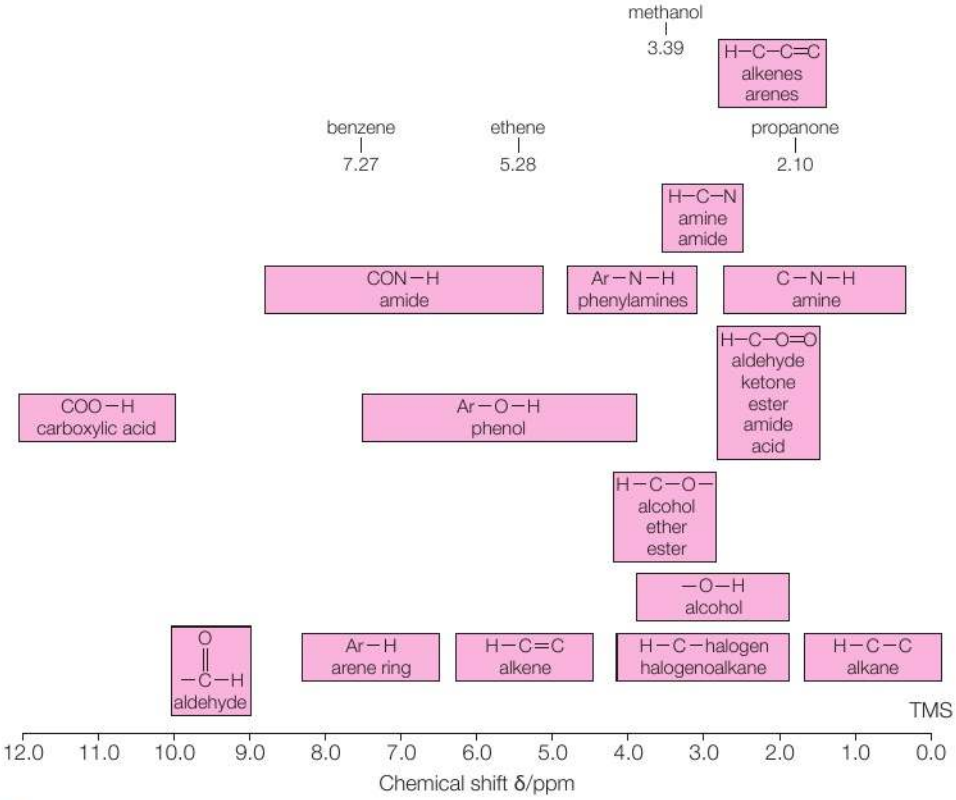
▲ **fig A** The low resolution ^{1}H NMR spectrum of propan-1-ol.



▲ **fig B** The low resolution ^{1}H NMR spectrum of propan-2-ol.

INTERPRETING CHEMICAL SHIFTS

To interpret chemical shifts you need a table of reference values with corresponding hydrogen atoms or a chart. **Fig C** shows a simplified version of the chart that will be provided in an examination, and you need to become familiar with using it.



▲ **fig C** ¹H NMR chemical shift ranges.

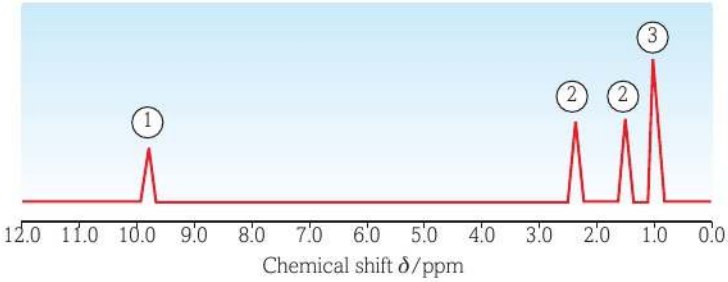
For two of the chemical shift ranges in this chart, the compounds included show trends in chemical shift values, as shown here in **table A**.

TYPE OF COMPOUND	TREND
halogenoalkane	$R_2CHF > R_2CHCl > R_2CHBr > R_2CHI$
alkane	$R_3CH > R_2CH_2 > RCH_3$

table A

BUTANAL AND BUTANONE

We will use the chart to interpret the low resolution ¹H NMR spectra of these isomers with the molecular formula C_4H_8O. **Fig D** shows the spectrum for butanal.



▲ **fig D** The low resolution ¹H NMR spectrum of butanal.

The peak at δ = 9.8 ppm is within the aldehyde range, so must be due to the H in the CHO group.

The peak at δ = 2.4 ppm is within the range for H—C—C=O, so must be due to the two H atoms in the CH_2 group joined to the CHO group.

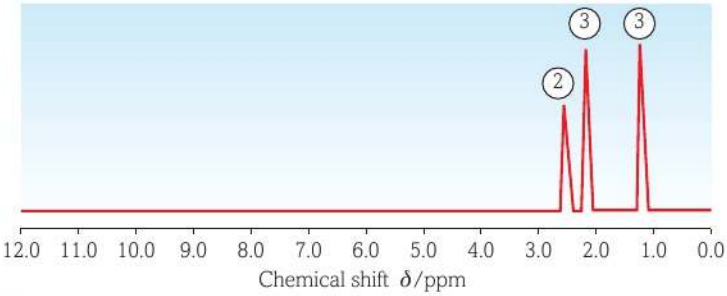
The peak at δ = 1.6 ppm is within the range for H—C—C (alkane), so must be due to the two H atoms in the CH_2 group not directly joined to the CHO group. Note that although the spectrum is for an aldehyde, the shift for an alkane is appropriate because this CH_2 group is part of the $CH_3CH_2CH_2$ alkyl group.

The peak at δ = 1.0 ppm is within the range for H—C—C (alkane), so must be due to the three H atoms in the CH_3 group.

EXAM HINT

Remember that the areas under the peaks (integrals) will give you the simplest ratios of the hydrogen environments. Ratios of 3:3:2 could mean hydrogen numbers of 6:6:4 in the actual molecule. You will need to check the molecular ion peak in the mass spectrum to help identify the molecule.

Now for the spectrum of butanone, shown in **Fig E**.



▲ **fig E** The low resolution ^{1}H NMR spectrum of butanone.

The peak at δ = 2.5 ppm is within the range for H—C—C=O, so must be due to the two H atoms in the CH_2 group joined to the CO group.

The peak at δ = 2.1 ppm is within the range for H—C—C=O, so must be due to the three H atoms in the CH_3 group joined to the CO group.

The peak at δ = 1.1 ppm is within the range for H—C—C (alkane), so must be due to the three H atoms in the CH_3 group not directly joined to the CO group.

LEARNING TIP

Both low resolution and high resolution (see **Topic 15E.8** for more details) ^{1}H NMR spectra indicate three things:

- The number of peaks represents the number of different chemical environments of the hydrogen atoms (protons).
- The areas under the peaks indicate the relative numbers of protons in each chemical environment.
- The chemical shift of a peak can indicate the functional group responsible (e.g. aldehyde or ketone).

CHECKPOINT

SKILLS ANALYSIS

1. Predict the number of peaks in the ^{1}H NMR spectrum of each of the isomers of C_5H_{12}.
2. Why does an integration trace not always indicate the actual numbers of hydrogen atoms in a molecule?
3. How would you easily decide from a ^{1}H NMR spectrum whether a compound is ethylamine ($CH_3CH_2NH_2$) or ethanamide (CH_3CONH_2)?

SUBJECT VOCABULARY

peak a peak in a ^{1}H NMR spectrum shows the presence of hydrogen atoms (protons) in a specific chemical environment

integration trace this shows the relative numbers of equivalent protons (i.e. in the same chemical environment)

15E 8 SPLITTING PATTERNS IN ^{1}H NMR SPECTRA

SPECIFICATION REFERENCE
15.20(iii) 15.20(iv)

LEARNING OBJECTIVES

- Be able to use both low and high resolution proton NMR spectroscopy to:
  (i) deduce the splitting patterns of adjacent, non-equivalent protons using the (n+1) rule and hence suggest the possible structures for a molecule
  (ii) predict the chemical shifts and splitting patterns of the 1H atoms in a given molecule.

HIGH RESOLUTION ^{1}H NMR SPECTROSCOPY

In **Section 15E.7** we introduced NMR by starting with the low resolution form, although only the high resolution form is actually used now. Both forms have two features in common:

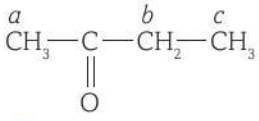
- Peaks at different chemical shift values help to identify the different chemical environments of the hydrogen atoms. If the environments are identical, then the protons are often described as being equivalent (i.e. they are **equivalent protons**).
- The relative peak areas (i.e. the ratio of peak areas) help to decide the numbers of hydrogen atoms responsible.

A high resolution ^{1}H NMR spectrum has one extra feature not seen in a low resolution spectrum – the peaks may have a **splitting pattern**. This means that the peak is split into a group of smaller peaks (sub-peaks) grouped very closely together. When considering these split peaks:

- you can take the chemical shift to be at the centre of the group of sub-peaks
- the relative peak areas are shown in the same way as in a low resolution spectrum – by an integration trace or by showing numbers next to the peaks
- remember that the numbers indicate the ratios of the numbers of protons – they may happen to be the actual numbers of protons, but this is not always the case.

WHAT CAUSES THE SPLITTING OF PEAKS?

Although the most important thing is for you to be able to interpret the splitting patterns, it will help if you understand how the splitting is caused. The fundamental point to be aware of is that hydrogen atoms (we will call them protons from now on) joined to a carbon atom have an influence on the protons on adjacent or neighbouring carbon atoms. Consider a molecule of butanone, in which the protons are labelled with letters:



▲ **fig A** The three different chemical environments for the protons in butanone.

The protons labelled *a* are not influenced by any other protons because there are none on the adjacent carbon atom. The protons labelled *b* are influenced by the protons labelled *c* and vice versa. These influences are what cause splitting of the peaks (sometimes referred to as spin–spin coupling), usually considered in terms of the n+1 rule. Very simply, if a carbon atom has n protons, then the peaks on adjacent carbon atoms are split into n+1 sub-peaks. The split peaks are sometimes referred to as **multiplets**, but more often by a term indicating the number of sub-peaks.

If the peak is not split, it is called a singlet.

If the peak is split into two sub-peaks, it is called a doublet.

If the peak is split into three sub-peaks, it is called a triplet.

If the peak is split into four sub-peaks, it is called a quartet.

The sub-peaks in a split peak have distinctive shapes, owing to splitting into different sub-peak areas, as shown in **table A**.

N	*N*+1	MULTIPLET	RATIO OF PEAK AREAS
0	1	singlet	1
1	2	doublet	1 : 1
2	3	triplet	1 : 2 : 1
3	4	quartet	1 : 3 : 3 : 1

table A

LEARNING TIP

The splitting pattern of a group of protons indicates nothing about the group of protons being split, but relates only to the number of protons on an adjacent atom.

UNDERSTANDING THE SPECTRUM OF BUTANONE

We can apply the information provided above to explain the spectrum of butanone in **table B**.

PROTONS	MULTIPLET	EXPLANATION
a	singlet	There are no protons on the adjacent carbon atom (C=O), so there is no splitting.
b	quartet	There are three protons on the adjacent CH_3 group, so applying the n+1 rule, the peak for the two protons is split into a quartet.
c	triplet	There are two protons on the adjacent CH_2 group, so applying the n+1 rule, the peak for the three protons is split into a triplet.

table B

TWO MORE POINTS TO REMEMBER

- We have said that protons cause splitting of peaks on adjacent atoms. While this is generally true, it does not apply in cases where the molecule is completely symmetrical. For example, there is splitting of peaks for 1,1-dichloroethane, $CHCl_2$—CH_3. The single proton splits the peak for the CH_3 group into a doublet, and the three protons in the CH_3 group split the CH peak into a quartet, following the n+1 rule. However, in 1,2-dichloroethane, CH_2Cl—CH_2Cl, neither of the CH_2 groups affects the protons of the other CH_2 group because all of the protons are equivalent (the CH_2 groups are in the same chemical environment).
- Now consider 1-chloropropane, CH_3—CH_2—CH_2—Cl. This time, the two CH_2 groups are in different chemical environments (so the protons are not equivalent). One is joined to CH_3 and CH_2, and the other is joined to CH_2 and Cl, so there will be splitting of peaks by the CH_2 groups. However, this example has been included to make a different point. The first CH_2 group would be influenced by the CH_3 group on its left and by the CH_2 group on its right. Using the n+1 rule, you would make the correct prediction that the CH_3 and the CH_2 group acting together would cause splitting into six sub-peaks – this splitting pattern would be described as a sextet. In other compounds, a quintet would be formed by four protons on adjacent carbon atoms – these could be two CH_2 groups or one CH_3 group and one proton.

PRACTICE MAKES PERFECT, AGAIN!

The best way to check how well you understand how to use ^{1}H NMR spectra is to try lots of examples. We will use three examples, all of which require you to use the chart of chemical shifts in **Section 15E.7**.

WORKED EXAMPLE 1

Predicting the appearance of a spectrum (chemical shifts and splitting patterns)
What would you predict about the appearance of the ^{1}H NMR spectrum of methyl propanoate ($CH_3CH_2COOCH_3$)?

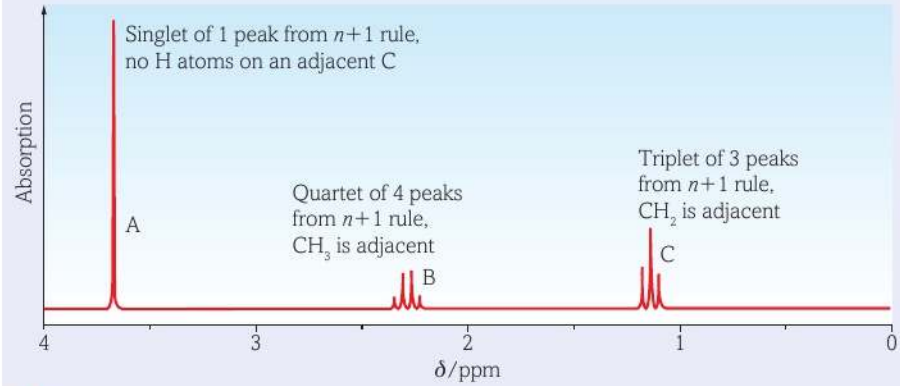
Answer

The CH_3 in the ethyl group is part of an alkyl group and is joined only to a hydrocarbon (CH_2) group, so it should have a peak in the range δ = 1.8–0.1 ppm, which is split into a triplet because of the two protons in the adjacent CH_2 group.

The CH_2 in the ethyl group is part of an alkyl group and is joined to a hydrocarbon (CH_3) group but also to the C of the COO (ester) group, so it should have a peak in the range δ = 2.9–1.8 ppm, which is split into a quartet because of the three protons in the adjacent CH_3 group.

The CH_3 in the methyl group is joined only to an oxygen atom in the COO group, so it should have a peak in the range δ = 4.2–2.8 ppm, which is a singlet. This peak is not split because there are no protons on the adjacent atom.

Fig B shows the actual spectrum of methyl propanoate, so you can check whether the predictions are correct.

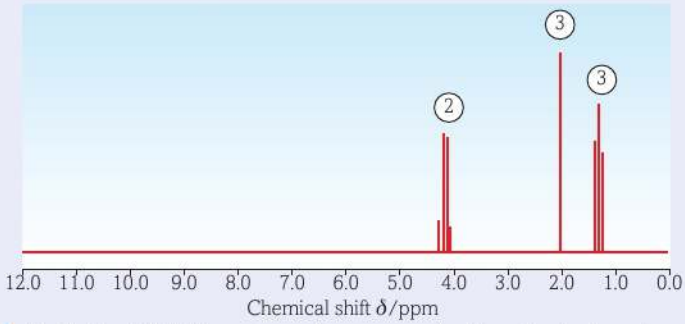


▲ **fig B** The ^{1}H NMR spectrum of methyl propanoate.

WORKED EXAMPLE 2

Predicting the structure of a compound from its spectrum

Fig C shows the ^{1}H NMR spectrum of a compound with the molecular formula $C_4H_8O_2$.



▲ **fig C** The ^{1}H NMR spectrum of a compound with molecular formula $C_4H_8O_2$.

What is its structure?

Remember that the spectrum shows four features:

- the number of different hydrogen environments
- relative peak areas
- chemical shifts
- splitting patterns

all of which need to be considered together.

Answer

There are three peaks, so there are three different hydrogen environments. The peak on the left is a quartet at δ = 4.1 ppm representing two protons. This suggests a CH_2 group next to a CH_3 group and an O in an alcohol, ether or ester, so $O{-}CH_2{-}CH_3$.

The peak in the middle is a singlet at δ = 2.0 ppm representing three protons. This suggests a CH_3 group joined to C=O in an aldehyde, ketone, ester, amide or acid, so $CH_3{-}CO{-}$.

The peak on the right is a triplet at δ = 1.3 ppm representing three protons. This suggests a CH_3 group next to a CH_2 group, so $-CH_2{-}CH_3$.

The final step is to put all these interpretations together. This is often easy to do but difficult to explain. The only structure that fits the interpretations is $CH_3{-}CO{-}O{-}CH_2{-}CH_3$, which is the ester ethyl ethanoate.

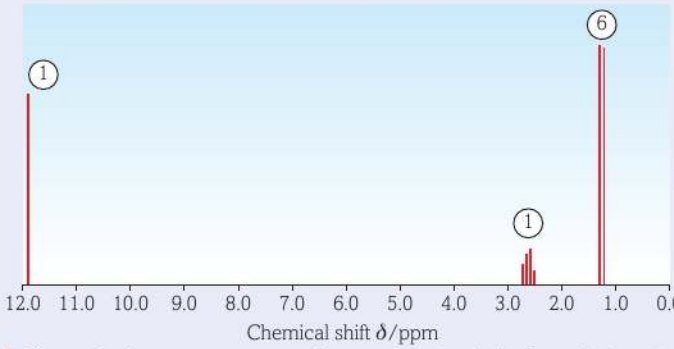
WORKED EXAMPLE 3

Another prediction of the structure of a compound from its spectrum

Fig D shows the ^{1}H NMR spectrum of a compound with the molecular formula $C_4H_8O_2$ (the same molecular formula as in Example 2).

Another prediction of the structure of a compound from its spectrum

Fig D shows the ^{1}H NMR spectrum of a compound with the molecular formula $C_4H_8O_2$ (the same molecular formula as in Example 2).



▲ **fig D** The ^{1}H NMR spectrum of a compound with the formula $C_4H_8O_2$.

What is its structure?

Answer

The peak on the left is a singlet at δ = 11.9 ppm representing one proton. This suggests a proton in a COOH group, so COOH.

The peak in the middle is a quartet (or multiplet) at δ = 2.6 ppm representing one proton. This suggests a CH group joined to a CH_3 group, so $CH_3{-}CH{-}$.

The peak on the right is a doublet at δ = 1.2 ppm representing six protons. This suggests two CH_3 groups next to a CH group, so $(CH_3)_2{-}CH{-}$.

The final step is to put all these interpretations together. Again, this is easy to do but difficult to explain. The only structure that fits the interpretations is $(CH_3)_2{-}CH{-}COOH$ – this is the acid methylpropanoic acid.

CHECKPOINT

SKILLS REASONING

1. A compound has the molecular formula C_4H_8O. Its ^{1}H NMR spectrum contains only three singlets, with chemical shift values of 4.0, 3.4 and 2.1 ppm. The integration trace shows the ratios 2 : 3 : 3. What is its structure?
2. A compound has the molecular formula $C_5H_{10}O_2$. Its ^{1}H NMR spectrum contains a singlet with a chemical shift of 2.3 ppm, a quartet with a chemical shift of 5.0 ppm and a doublet with a chemical shift of 1.2 ppm. The integration trace shows the ratios 3 : 1 : 6. What is its structure?

SUBJECT VOCABULARY

equivalent protons hydrogen atoms in the same chemical environment

splitting pattern the appearance of a peak as a small number of small sub-peaks very close to each other

multiplets the different splitting patterns observed (singlets, doublets, triplets or quartets) in a high resolution ^{1}H NMR spectrum